

Josh Sar-Shalom

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EDUCATION

University of Central Florida (UCF) – Orlando, FL

Expected Graduation: Fall, 2027

- *B.S. in Computer Engineering*

GPA: 3.95

Honors/Awards: UCF SMART 2025 Certificate | Dean's List S26, F25, S24, F23 | AICE Diploma

Relevant Coursework: Digital Systems, Computer Organization, Computer Architecture, Linear Circuits II

WORK EXPERIENCE

TLC Engineering Solutions – *Smart Buildings/IoT Intern*

May 2026 – Present

- Prototyped IoT systems with ESP32-C6 microcontrollers and Sensirion SEN66 environmental sensors
- Led monthly stakeholder meetings to present updates and gather feedback on AI-driven Excel automation agents
- Migrated automation from cloud-based to local executable to reduce full-day manual Excel comparison workload
- Integrated SharePoint (SPFx) frontend components to streamline and modernize manual project management workflows

Semiconductor Manufacturing and Advances in Research Technology (UCF SMART 2025) – *Participant*

Jun 2025

- Gained hands-on experience on lithography, deposition, and etching machines to process UCF-patterned silicon wafers
- Performed hands-on assembly and testing of a Schottky diode using gold wire bonding and SMU instrumentation
- Engaged in cleanroom safety procedures directly from UCF research students and professors in cleanrooms and labs

PROJECTS

Bus Bar Attraction PLC Simulation – *Personal Project*

Jun 2026

- Modeled a 6-section bus bar track layout to execute block-zone control logic inside Beckhoff's TwinCAT 3
- Programmed an 18-rung ladder diagram for real-time vehicle tracking and collision prevention across 6 track sections
- Mapped track occupancy and vehicle status a dynamic HMI for real-time system visualization
- Simulated vehicle movement using TON timers to validate block-zone control and position tracking logic

ESP32 Webserver Light Toggle – *Personal Project*

May 2026 – Present

- Prototyped and tested ESP32 circuit components including servo control and 5 V power system
- Implemented webserver hosting for wireless interface for remote device control over a local network
- Developing PCB in KiCad and light switch enclosure in Fusion360 for ESP32 and servo for a low-profile design
- Integrated a modular design to enable future IoT smart home modules for expansions

NAND-Based Digital Logic Rom-Pixel-Display – *Personal Project*

Mar 2025 – Apr 2025

- Implemented a digital CPU to render pixel data using custom NAND-built logic gates with an addressable ROM chip
- Utilized SDL2 library in C to convert a 16x16 canvas to paste-ready, formatted, 16-bit, opcode instructions
- Built a foundational understanding of low-level hardware design through logic gate and memory integration
- Debugged RAM output timing issues, synchronizing rising and falling edge signals for proper display output

INVOLVEMENT

NASA L'Space Mission Concept Academy – *Lead Command & Data Hardware Engineer*

Sep 2024 – Dec 2024

- Tasked to plan, design, budget and organize a moon pit exploration mission for future human operations
- Constructed a software architecture flowchart detailing data processing, mission control, and subsystem interactions
- Designed a functional block diagram showcasing data flow, power distribution, and communication protocols
- Authored NASA standard MCR, SRR, MDR, and PDR documents to present team progress, designs, and requirements

Theta Tau, Rho Gamma Chapter – *Academic Board & Developers Committee Member*

Dec 2024 – Present

- Contributed to the updated Rho Gamma Theta Tau website design with Developers Committee
- Engaging with academic board members to host study ragers, internship applyathons and scholarship announcements
- Participated in community service and philanthropic events outreached by board members and local partners

SKILLS

Tools & Software: OpenPLC, TwinCAT 3 (Beckhoff), SolidWorks, Fusion, Siemens NX, Microsoft Office, CoPilot Studio

Programming & Development: C/C++, Python, Ladder Logic, MIPS, Vim, Git, Raspberry Pi, Esp32, Arduino IDE

Hardware & Lab Experience: Soldering, Gold Wire Bonding, KLayout, MDR/SSR/MCR documentation